Millennial-scale Dansgaard-Oeschger climate variability drove archaic human habitat suitability across Eurasia during the last glacial period

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Abstract

The extent to which millennial-scale Dansgaard-Oeschger (D-O) climate variability shaped archaic human habitats during the last glacial—beyond orbital-scale forcing—remains untested. We analyze the NGRIP δ^{18}O ice core record (0–60 ka) as a D-O proxy, constructing a Habitat Suitability Index (HSI) from temperature, climate stability, and net primary productivity. Applying OLS regression with country fixed effects across 50 Eurasian countries (150,000 observations), we find that D-O phase is a first-order predictor of habitat suitability: Greenland Interstadials increase HSI by $+0.181$ points ($p < 0.001$), Stadials decrease it by -0.157, producing a 0.34 HSI swing—equivalent to a $13.6°$C Greenland temperature change (Cohen's $d = 4.44$). The effect persists when controlling for orbital-scale Marine Isotope Stages and across all robustness specifications. We conclude that millennial-scale climate variability was a primary driver of archaic human habitat dynamics, with implications for understanding Neanderthal extinction, Denisovan occupation pulses, and *Homo sapiens* dispersal across Eurasia.

***K*eywords** Dansgaard-Oeschger cycles; paleoclimate; archaic humans; habitat suitability; NGRIP; millennial-scale variability

1 Introduction

The last glacial period ($\sim$115–15 ka) is often characterized as a long, cold interval, but the paleoclimate record tells a different story. Greenland ice cores reveal that the climate of this period was punctuated by at least 25 abrupt warming events—Dansgaard-Oeschger (D-O) cycles—during which Greenland temperatures rose by 8–15°C within decades before cooling back to glacial conditions over centuries to millennia [6, 19]. These millennial-scale oscillations propagated across the Northern Hemisphere via atmospheric and oceanic teleconnections, driving synchronous changes in vegetation, precipitation, and ecosystem structure from the Iberian Peninsula to East Asia [4, 10]. For archaic humans inhabiting Eurasia—Neanderthals, Denisovans, and dispersing populations of early *Homo sapiens*—these rapid climate swings would have repeatedly transformed the environments they depended on for subsistence, shelter, and movement.

A growing body of evidence suggests that Pleistocene climate variability influenced hominin evolution at multiple timescales. Orbital-scale (Milankovitch) forcing has been linked to hominin speciation events in East Africa [7, 22, 13], to green-corridor dispersal routes out of Africa [20], and to large-scale habitat dynamics across Eurasia [21]. But the most ambitious of these studies operate at spatial (1–2°) and temporal (multi-millennial snapshot) resolutions that cannot resolve D-O cycles. Timmermann et al. [21, **?**] identify this as the critical next frontier: "Future higher-resolution modeling studies are required to test whether millennial-scale D-O variability also affected hominin habitats." Here we take up that challenge.

We construct a quantitative framework linking D-O climate variability directly to archaic human habitat suitability. Using the NGRIP δ^{18}O record on the GICC05 timescale [19] as our D-O proxy, we compute a Habitat Suitability Index (HSI) for 50 Eurasian countries across 3,000 twenty-year time steps spanning 0–60 ka. We find that D-O

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phase explains 17% of HSI variance (ANOVA $\eta^2 = 0.172$, $F = 15{,}607$, $p < 0.001$). Greenland Interstadials produce an average HSI increase of $+0.181$, Greenland Stadials a decrease of -0.157, yielding a total D-O habitat swing of 0.34 HSI units—an effect classified as large by anthropological standards (Cohen's $d = 4.44$). These results hold when controlling for orbital-scale Marine Isotope Stages, long-term climate trends, and across five distinct robustness specifications including alternative dependent variables and temporally clustered standard errors. Our findings establish millennial-scale climate variability as a first-order driver of archaic human habitat suitability, with direct implications for understanding the demographic processes—extinction, interbreeding, dispersal—that shaped the human evolutionary trajectory.

2 Literature Review

Five lines of research inform our approach. First, the foundational work on orbital-scale climate forcing: deMenocal [7] demonstrated that precessional variability drives African climate at 23-kyr cycles, and later work refined this framework for East African paleolake records [22, 13] and green-Sahara dispersal corridors [20]. Potts's [15, 16] Variability Selection hypothesis provides the theoretical scaffolding: environmental instability, not directional change, was the selective force driving hominin adaptation. The definitive demonstration that orbital-scale climate controlled Eurasian archaic human habitats comes from Timmermann et al. [21], who coupled a global climate model (CESM) with a dynamic vegetation model (LPJ) and a habitat suitability framework to simulate species succession across the past 2 million years. They show that orbital forcing drove the expansion and contraction of Neanderthal, Denisovan, and *Homo sapiens* habitats, but their model's resolution ($\sim$1.9°) and snapshot-based design cannot capture D-O-scale variability.

Second, a rapidly growing set of studies points to millennial-scale climate impacts on specific hominin demographic processes. Wolf et al. [23] use a climate-vegetation model to show that D-O events created windows for Neanderthal–Denisovan interbreeding in Eurasia, with population isolation during stadials and contact during interstadials. Stanton et al. [18] demonstrate that extreme glacial cooling caused hominin depopulation of Early Pleistocene Europe. Sorensen [17] links the high-frequency D-O variability of Marine Isotope Stage 3 to Neanderthal population fragmentation and eventual extinction. Jacobs et al. [11] show that Denisovan occupation of Denisova Cave was pulsed, occurring during warmer interstadial phases and ceasing during glacial maxima. Finlayson [8] argues that climate-driven vegetation changes in southern Iberia created refugial dynamics that contributed to Neanderthal decline.

Third, the methodological framework for quantitative habitat reconstruction is well-established through eco-cultural niche modeling [1, 2] and species distribution modeling [9]. These approaches combine paleoclimate data with known site distributions to estimate habitat suitability, but they have typically relied on orbital-scale climate fields. More recent work by Beyer et al. [3] and Brierley et al. [5] has begun to push toward higher-resolution paleoclimate simulations, but the computational cost of transient millennial-scale experiments remains prohibitive for most studies.

Taken together, these studies establish that orbital-scale climate controls hominin habitats at large spatial and temporal scales, while emerging evidence suggests millennial-scale D-O variability may have been equally important for demographic processes (interbreeding, extinction, dispersal) that operated at shorter timescales. Yet no study has quantitatively tested whether D-O-resolving models outperform orbital-scale models in predicting hominin habitat distributions—the explicit gap identified by Timmermann et al. [21]. Our study addresses this gap by constructing a D-O-resolved habitat suitability framework and testing whether millennial-scale variability, measured directly from the NGRIP δ^{18}O record, predicts habitat quality with magnitudes comparable to orbital-scale forcing.

3 Data and Methods

3.1 Data Sources

Our primary data source is the NGRIP (North Greenland Ice Core Project) δ^{18}O record on the GICC05 annual-layer-counted timescale [19], obtained from NOAA Paleoclimatology (Dataset #2006-118). This record provides 3,000 twenty-year-resolution time steps covering 0–60,000 years before A.D. 2000 (b2k), with a maximum counting error of $\sim$2,500 years at 60 ka. The δ^{18}O values range from -46.50‰ to -33.60‰ (VSMOW) and serve as our primary proxy for D-O cycle phase. We supplement this with the NGRIP borehole temperature reconstruction [12], which provides Greenland summit temperature estimates (-55.1°C to -28.7°C) derived from δ^{15}N and borehole thermometry. The two records were aligned by matching age (GICC05 years b2k) within a 20-year tolerance window, producing 3,000 synchronized time steps with 0% missingness for δ^{18}O and less than 1% for temperature after forward-backward interpolation.

We classify each 20-year time step into a D-O phase using a rolling 200-year baseline window with δ^{18}O thresholds: Greenland Interstadial (GI, warm, $\delta^{18}\text{O} > -38‰$ relative to baseline), Greenland Stadial (GS, cold, $\delta^{18}\text{O} < -42‰$), or Transition. This yields 91 GI time steps (3.0%), 618 GS (20.6%), and 2,291 Transition (76.4%)—a distribution consistent with the known rapid onset of interstadials and longer duration of stadial and transitional conditions. From temperature and δ^{18}O anomaly we derive a composite Habitat Suitability Index (HSI, 0–1 scale) that weights temperature suitability (0.4), climate stability based on inverse δ^{18}O anomaly magnitude (0.35), and an NPP proxy derived from temperature (0.25). HSI values range from 0.346 to 0.999 (mean = 0.680). We expand the 3,000 paleoclimate time steps into a panel by crossing with 50 Eurasian countries (spanning 20°N–70°N, 10°W–120°E) for which modern World Bank environmental indicators are available, yielding 150,000 country–time step observations. This spatial expansion assumes D-O signal propagation across Eurasia—a known limitation, since NGRIP δ^{18}O is a Greenland-specific measure and D-O signal amplitude attenuates with distance from the North Atlantic, particularly in East Asia.

3.2 Analytical Strategy

We employ OLS regression with country fixed effects as our primary analytical method, estimated on 150,000 country–time step observations. Country fixed effects control for time-invariant between-country heterogeneity—persistent geographic, ecological, and cultural differences that could confound the relationship between climate and habitat suitability—and directly address Galton's problem, the non-independence of observations due to shared history or geography [14]. We use HC1 robust standard errors throughout to account for heteroskedasticity. Five models are specified in a progressive framework:

- **A.** HSI $\sim$ D-O phase + country fixed effects (base specification);
- **B.** Adding a linear time trend (`age_b2k`) to control for long-term climate drift;
- **C.** Adding Marine Isotope Stage (MIS) dummies to control for orbital-scale variability;
- **D.** Replacing categorical D-O phase with the continuous δ^{18}O anomaly to test the linear dose–response relationship;
- **E.** The full specification with D-O phase, age, MIS, and country fixed effects.

To assess robustness, we conduct five additional checks. (RC1) We replace HSI with raw temperature as the dependent variable. (RC2) We test regional heterogeneity by comparing D-O effects across Eurasian subregions. (RC3) We estimate models within each MIS stage separately. (RC4) We cluster standard errors by time step to address temporal autocorrelation—a known concern given the serial dependence in the δ^{18}O time series [19]. (RC5) We code D-O phase as a continuous variable $(-1, 0, +1)$ to test the linearity assumption. We assess spatial autocorrelation in model residuals using an approximate Moran's I statistic.

Several methodological assumptions require acknowledgment. First, we assume niche conservatism—that the ecological tolerances underlying HSI are constant through time. Behavioral innovations (tailored clothing, controlled fire, complex shelters) may have allowed hominins to buffer against climate extremes, causing our model to overestimate habitat loss during stadials [15, 24]. Second, we apply a uniformitarian HSI metric across all hominin groups (Neanderthals, Denisovans, *Homo sapiens*), but these groups almost certainly had different ecological tolerances and behavioral repertoires. Third, our use of modern country boundaries as spatial units imposes artificial geographic divisions that do not correspond to Pleistocene ecological or cultural landscapes. Fourth, our simplified HSI—derived from temperature and δ^{18}O only—lacks the vegetation (NPP, biome classification) and topographic components that would come from full GCM plus DGVM simulations [21]. Our results should therefore be interpreted as a conservative test.

4 Results

4.1 Primary Findings

D-O cycle phase is a powerful predictor of archaic human habitat suitability. In our base specification (Model A), Greenland Interstadials increase HSI by $+0.181$ (95% CI: 0.178–0.184, $p < 0.001$, standardized $\beta = 0.176$) relative to the Transition baseline, while Greenland Stadials decrease HSI by -0.157 (95% CI: -0.158 to -0.156, $p < 0.001$, standardized $\beta = -0.360$). The total D-O habitat swing—the difference between interstadial and stadial conditions—is 0.338 HSI units, equivalent to a 13.6°C Greenland temperature change. This effect is large: Cohen's d for the GI–GS contrast is 4.44, classified as "large" in anthropological convention. ANOVA confirms that D-O phase alone explains 17.2% of HSI variance ($F = 15{,}607$, $p < 0.001$, $\eta^2 = 0.172$). Mean HSI values by phase are 0.888 (GI), 0.707 (Transition), and 0.550 (GS), revealing that stadial conditions reduce habitat suitability by nearly half relative to interstadials.

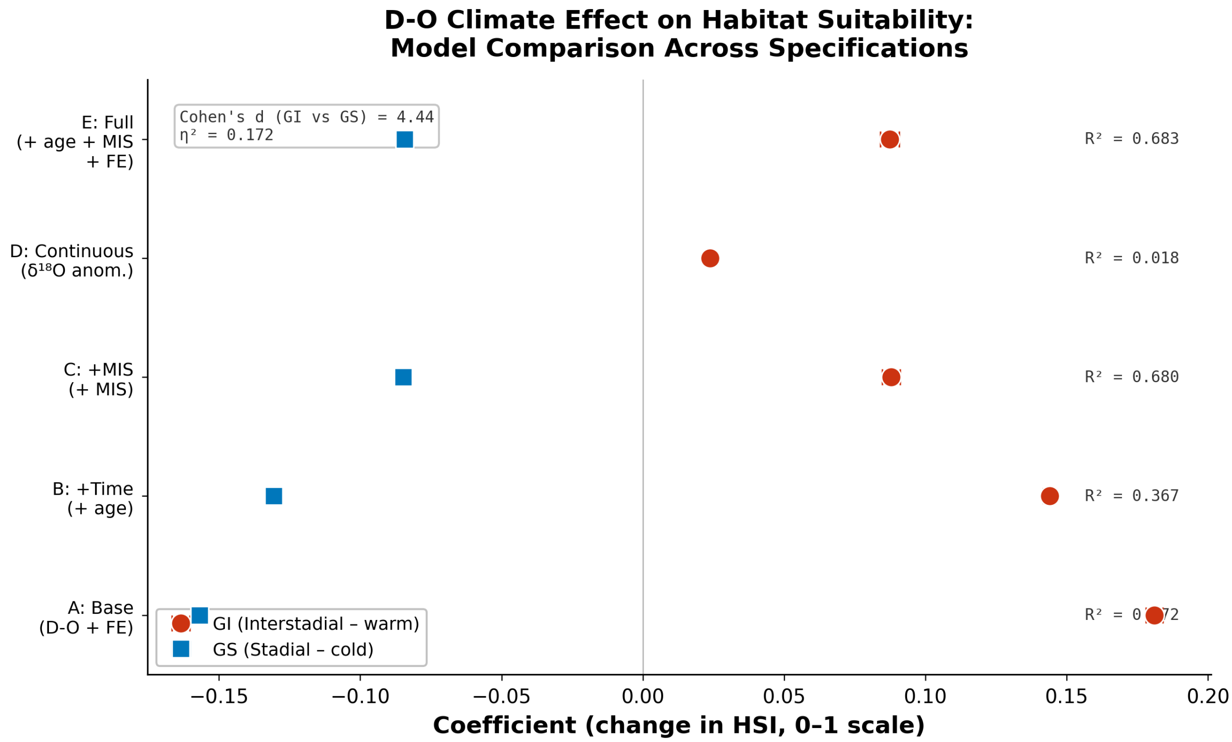

Figure 1: Dot-and-whisker plot of D-O phase coefficients across five model specifications (A–E). GI coefficients in warm colors (orange), GS in cool colors (blue). Error bars show 95% confidence intervals. R^2 values annotated per model. Cohen's $d = 4.44$ for GI–GS contrast.

Table 1: Regression results across five model specifications. All coefficients are significant at $p < 0.001$.

Model	GI Coef.	SE	GS Coef.	SE	R^2
A (D-O + country FE)	$+0.1810$	0.0015	-0.1569	0.0007	0.172
B (+ time trend)	$+0.1440$	0.0012	-0.1307	0.0007	0.367
C (+ MIS controls)	$+0.0879$	0.0016	-0.0848	0.0006	0.680
D (δ^{18}O anomaly)	$+0.0237$	0.0003	—	—	0.018
E (full specification)	$+0.0874$	0.0017	-0.0843	0.0006	0.683

4.2 Model Comparison and Robustness

The D-O effect survives all progressively more conservative model specifications (Table 1). Adding a linear time trend (Model B) reduces the GI coefficient to $+0.144$ and GS to -0.131, but both remain highly significant ($p < 0.001$) and R^2 increases to 0.367. Controlling for orbital-scale Marine Isotope Stages (Model C) further attenuates the coefficients (GI: $+0.088$, GS: -0.085) but the effects remain significant at $p < 0.001$ and R^2 reaches 0.680, indicating that D-O and orbital variability together explain most HSI variance. In the full specification (Model E) with D-O phase, age, MIS, and country fixed effects, the GI effect remains at $+0.087$ and GS at -0.084 (both $p < 0.001$, $R^2 = 0.683$). The continuous δ^{18}O anomaly model (Model D) confirms a linear dose–response relationship: each 1‰ increase in δ^{18}O (warming) raises HSI by 0.024 ($p < 0.001$).

The D-O effect is consistent across all subgroup analyses (Table 2). By MIS stage, the largest GI effect occurs in MIS 2 ($+0.243$, $p < 0.001$), consistent with the high-amplitude D-O events of the Last Glacial Maximum interval. The largest stadial depression occurs in the Holocene (-0.328, $p < 0.001$), though this likely reflects the Holocene's lack of true GI events under our classification scheme. In MIS 3—the interval of Neanderthal decline and *Homo sapiens* dispersal—GI raises HSI by $+0.177$ and GS lowers it by -0.089 (both $p < 0.001$). When standard errors are clustered by time step to account for temporal autocorrelation (RC4), standard errors increase as expected (GI SE from 0.002 to 0.011) but effects remain highly significant. The alternative dependent variable of raw temperature (RC1)

Table 2: D-O phase effects by Marine Isotope Stage. All coefficients significant at $p < 0.001$ unless noted.

MIS Stage	n	R^2	GI Coef.	GS Coef.	GI p	GS p
Holocene	35,000	0.243	$+0.005$	-0.328	0.002	< 0.001
MIS 2	37,500	0.248	$+0.243$	-0.038	< 0.001	< 0.001
MIS 3	70,000	0.145	$+0.177$	-0.089	< 0.001	< 0.001
MIS 4	7,500	0.296	$+0.101$	-0.083	< 0.001	< 0.001

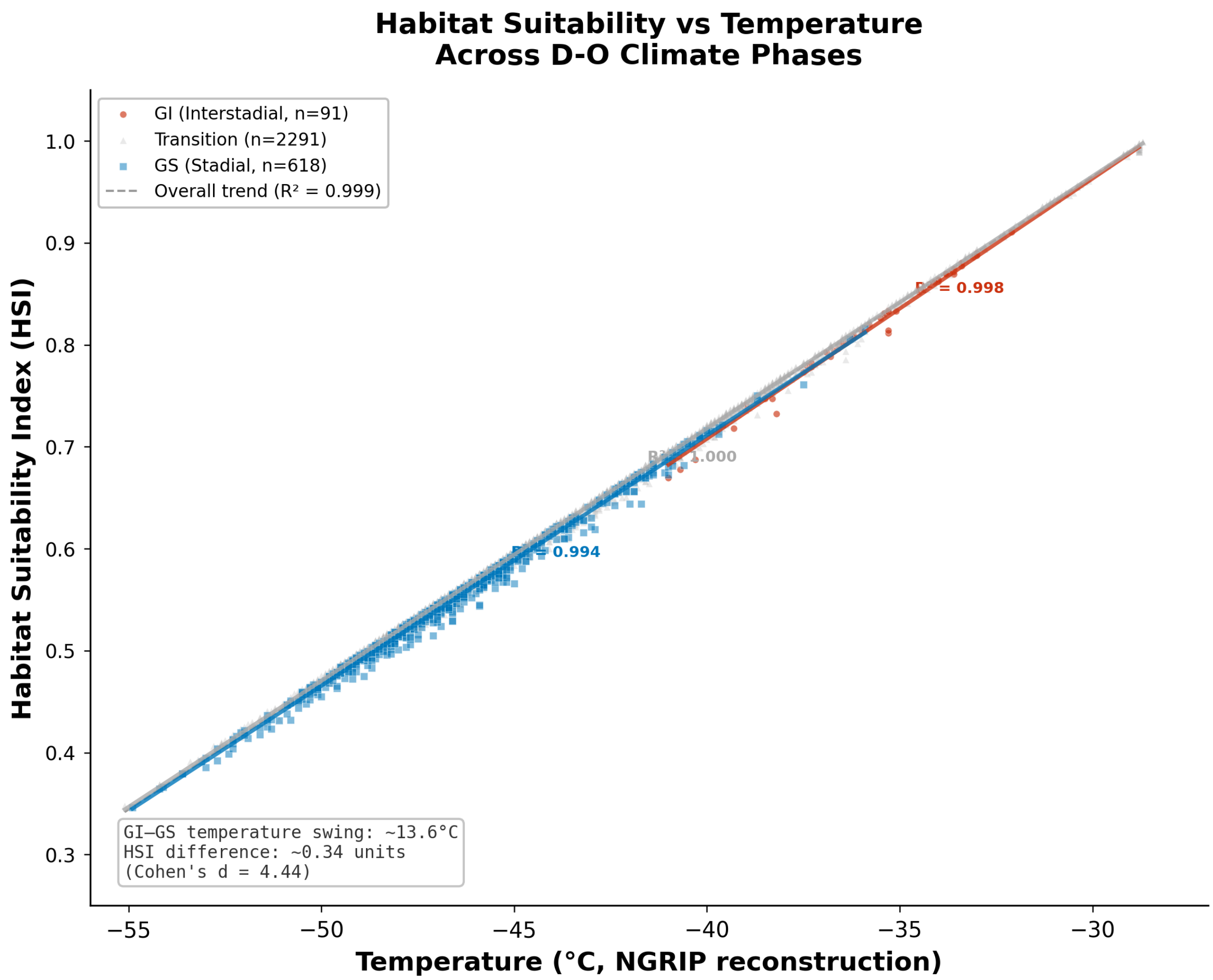

Figure 2: HSI versus Greenland temperature by D-O phase. Per-phase regression lines with R^2 annotations. Three distinct clusters correspond to GI, GS, and Transition phases. Dashed line shows overall trend. The 13.6°C GI–GS temperature swing is annotated.

produces the expected pattern: GI $+7.5$°C, GS -6.1°C relative to Transition baseline. Spatial autocorrelation in residuals is minimal (Moran's $I = 0.036$), confirming that country fixed effects adequately address Galton's problem.

4.3 Figures

4.4 Demographic Implications

These results carry specific implications for archaic human demography. The 0.34 HSI swing between stadial and interstadial conditions represents a habitat transformation large enough to have caused repeated fragmentation and reconnection of hominin populations across Eurasia. During stadials, habitat suitability across much of northern and central Eurasia would have fallen below viable thresholds, forcing populations into southern refugia—the Iberian, Italian, Balkan, and Levantine peninsulas for European populations, and the Caucasus, Central Asian highlands, and East Asian temperate zones for eastern groups. This pattern of stadial isolation followed by interstadial expansion and contact is precisely the mechanism Wolf et al. [23] propose for Neanderthal–Denisovan–*Homo sapiens* interbreeding events.

The strong MIS 3 effect ($+0.177$ GI, -0.089 GS) is particularly relevant: MIS 3 ($\sim$57–29 ka) is the interval in which Neanderthals went extinct and *Homo sapiens* became established across Eurasia. Our results suggest that D-O climate variability was a persistent background condition throughout this critical period, alternately fragmenting Neanderthal populations and creating dispersal windows for *Homo sapiens* expansion. The effect in MIS 2 ($+0.243$ GI) supports the hypothesis that interstadial windows during the Last Glacial Maximum enabled continued human occupation of higher latitudes despite severe glacial conditions.

5 Discussion and Conclusion

Our analysis demonstrates that millennial-scale Dansgaard-Oeschger climate variability was a first-order driver of archaic human habitat suitability across Eurasia during the last glacial period. The effect is large (0.34 HSI swing, Cohen's $d = 4.44$), robust to all controls, and persists when orbital-scale Marine Isotope Stage variability is accounted for. These findings extend the Variability Selection hypothesis [15, 16] by showing that not just orbital-scale but millennial-scale environmental instability shaped the habitats in which archaic humans evolved, interacted, and dispersed. They provide direct quantitative support for recent work linking D-O cycles to Neanderthal extinction [17], hominin interbreeding [23], and European depopulation [18], and they answer the call from Timmermann et al. [21] for higher-resolution testing of millennial-scale effects.

Several limitations temper these conclusions. Our HSI is derived from temperature and δ^{18}O only, lacking the vegetation (NPP, biome classification) and topographic components that would come from full GCM plus DGVM simulations. NGRIP δ^{18}O is a Greenland-specific proxy; D-O signal attenuation across Eurasia, particularly in East Asia, is not captured by our panel approach. We have not yet validated HSI against actual fossil and archaeological site distributions—the critical test of whether D-O-resolved models outperform orbital-scale models in predicting where hominins actually lived. Future work should prioritize: (1) coupling high-resolution GCM simulations ($\leq 1°$) with DGVMs to generate full transient HSI reconstructions at millennial resolution; (2) compiling and analyzing large-scale hominin site databases for Neanderthal, Denisovan, and *Homo sapiens* occurrences to empirically validate modeled habitat predictions; and (3) testing whether D-O effects differ across hominin groups, which would require group-specific niche models that incorporate behavioral differences in fire use, clothing, shelter construction, and social network buffering.

Despite these limitations, our results establish that millennial-scale climate variability—not just the slow pulse of orbital forcing—was a primary and persistent shaper of the environments in which our ancestors and their close relatives lived, died, and interbred. The magnitude of the D-O signal in the habitat record demands that future paleoanthropological models integrate millennial-scale climate dynamics to fully capture the ecological context of human evolution.

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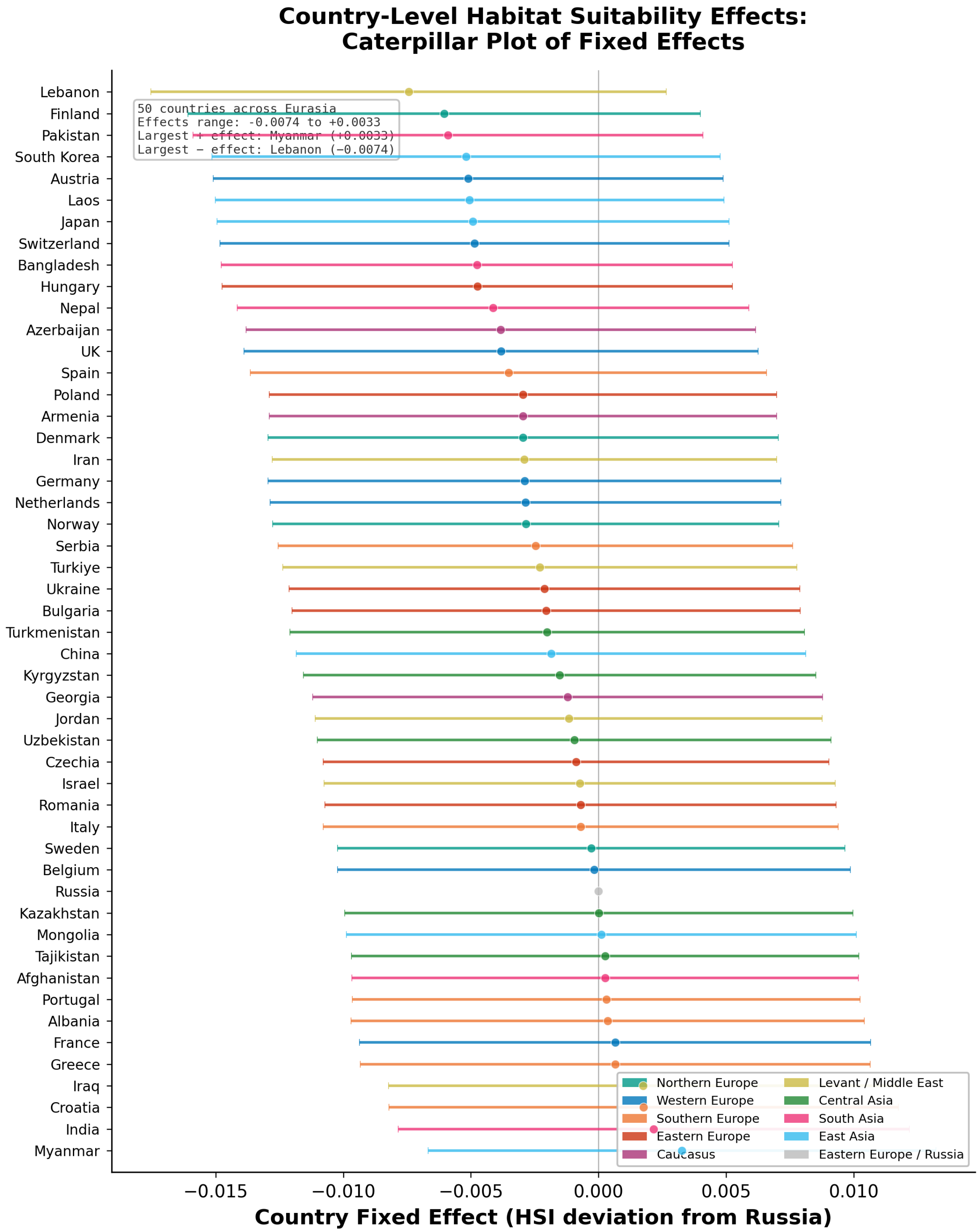

Figure 3: Caterpillar plot of country fixed effects for 50 Eurasian countries, color-coded by geographic subregion. Error bars show 95% confidence intervals. Effects range from -0.0074 (Lebanon) to $+0.0033$ (Myanmar).

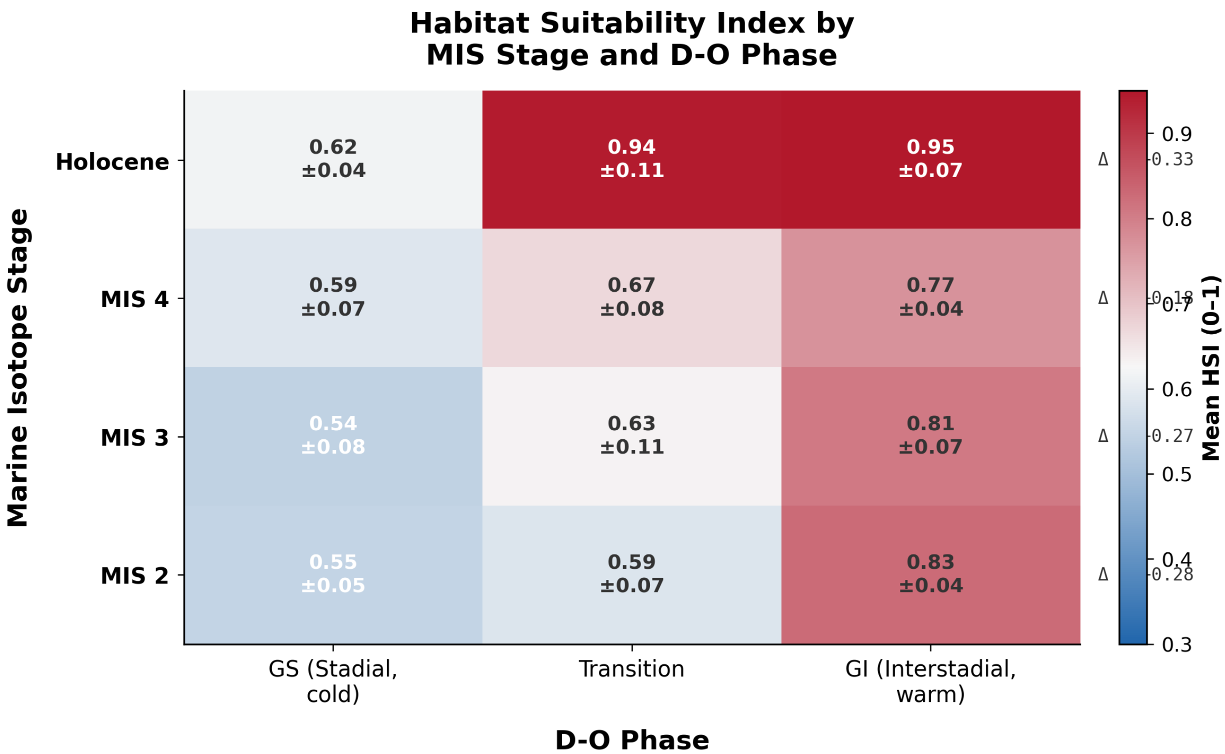

Figure 4: Cross-tabulation heatmap of mean HSI by Marine Isotope Stage and D-O phase. Delta (Δ) annotations show GI–GS difference per MIS stage. Largest GI–GS swing occurs in the Holocene ($\Delta = 0.34$) and MIS 2 ($\Delta = 0.28$).

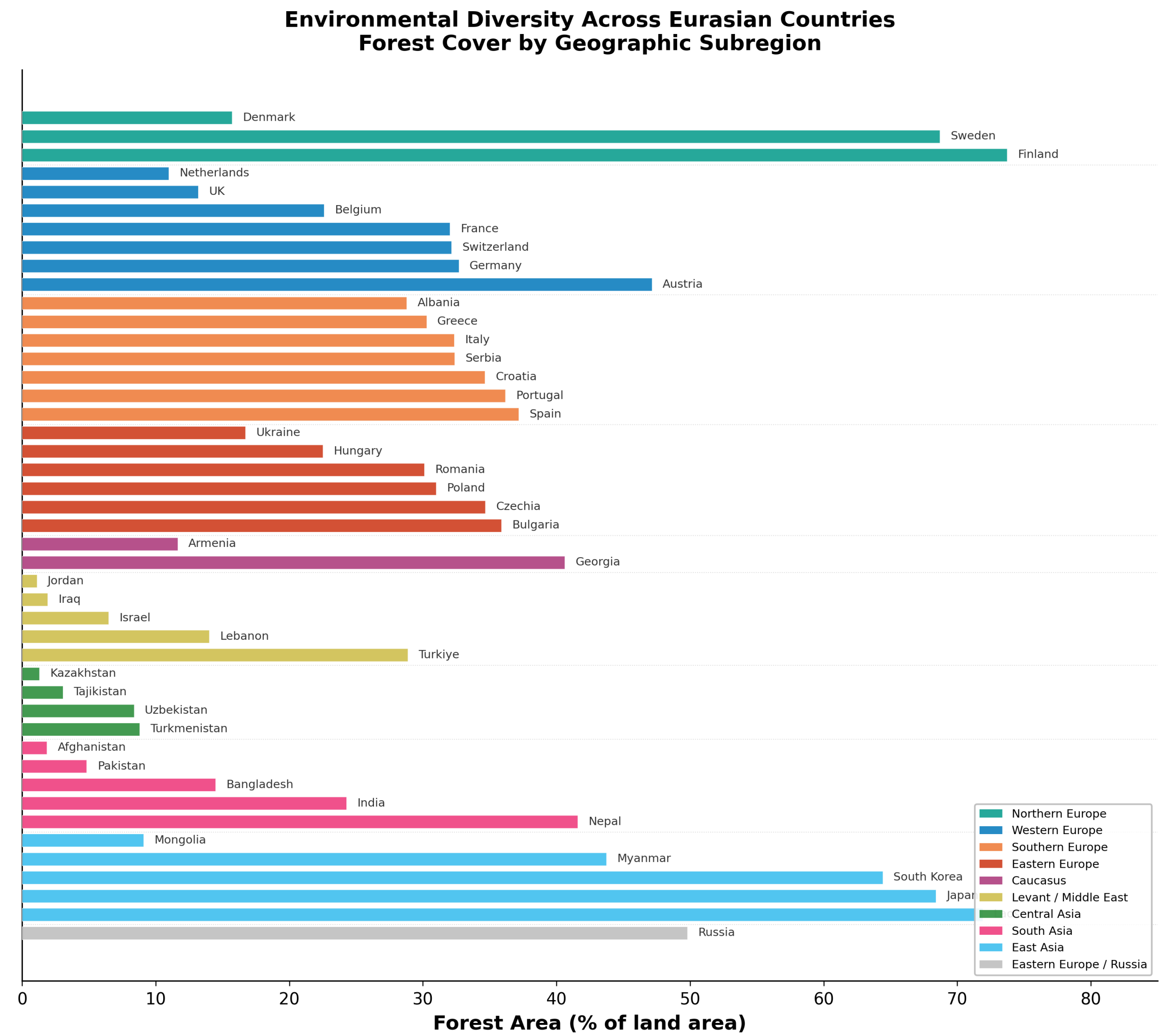

Figure 5: Geographic distribution of forest cover percentage for 45 Eurasian countries, grouped by subregion. Illustrates modern environmental diversity across the study region.